Rising Skill, Falling Skill: Why a Single Baseline Cannot Tell You Whether an Ocean Forecast Model Has Learned Anything

Deepak Krishna

MarisAI, India
deepakkrishna2206@gmail.com

Abstract. Statistical forecasts of ocean state are conventionally scored against *persistence*, the last observed value. We show this single baseline is not merely incomplete but actively misleading, because it fails in a direction that flatters the model. Persistence decays with forecast horizon while a seasonal cycle does not, so the persistence-referenced score of a model that has learned nothing but the time of year stays flat, or rises, while the model genuinely degrades. We audit 13 operational ocean variables across 24 regime-stratified sites and four horizons (1 d to 30 d), using one gradient-boosted framework with no variable-specific code, and score every model against both persistence and a leakage-controlled day-of-year climatology. Median skill against persistence is 0.272 at 1 d and 0.273 at 30 d — essentially unchanged, which reads as a horizon-invariant forecaster. Over the same range median skill against climatology falls monotonically from 0.958 to -0.023, declining with horizon for 13 of 13 variables without exception; by 30 d 7 of 13 variables lose to the seasonal cycle outright, against 1 that lose to persistence. The two baselines therefore support opposite conclusions about the same predictions, and 11 model–horizon combinations beat persistence while losing to climatology. A three-arm feature ablation then shows that the natural explanation for those cases is wrong: 10 of the 11 retain their skill when every calendar feature is removed, while the most seasonally dependent variable in the set comfortably *beats* climatology. Losing to climatology therefore reports the baseline's strength, not the model's mechanism; diagnosing the latter needs an ablation. We further test whether these pooled models transfer to locations they never saw, using leave-one-site-out validation blocked in space and time simultaneously (1248 held-out fits). Transfer is only partial: median held-out skill at 7 d falls to 0.050 from 0.151 pooled — a roughly threefold drop — with 56.4% of held-out sites retaining positive skill. We report 5 model–horizon combinations that lose to persistence, and argue that reporting both baselines across multiple horizons should be a minimum standard for this class of work.

Keywords: Ocean forecasting · Forecast verification · Skill scores · Baseline selection · Spatial cross-validation · Gradient boosting

1 Introduction

An ocean forecast is useful only relative to the cheapest thing one could have done instead. In operational oceanography that reference is almost always *persistence*: predict that tomorrow equals today. Persistence is a famously hard baseline on autocorrelated geophysical series, and a model that beats it is normally taken to have learned something.

This paper argues that persistence alone cannot support that inference, and that the way it fails is systematically misleading rather than merely noisy.

The argument is structural. Persistence error grows with the forecast horizon, because the ocean decorrelates from its present state. The error of a *seasonal climatology* does not: knowing that it is February is exactly as informative thirty days ahead as it is one day ahead. A skill score computed against persistence therefore has a denominator that inflates with horizon, which buoys the score of a model whose own error is growing. A model that has learned nothing beyond the annual cycle can hold its skill against persistence flat, or even increase it, while genuinely degrading — and flat or rising skill with lead time is precisely what a reader interprets as long-range predictive content.

We encountered this pattern empirically. In a framework trained across 13 ocean variables, skill against persistence rose with horizon for 7 of them and, in the median, was no lower at thirty days than at one. Whether that reflected real long-range skill or a seasonal artefact was, with persistence as the only baseline, unanswerable. This paper resolves it by adding the baseline that discriminates, and the answer is largely the second: skill against climatology falls with horizon for 13 of 13 variables, and 11 model–horizon combinations beat persistence while losing to climatology outright.

1.1 Contributions

1. A **two-baseline skill audit** over 13 operational ocean variables × 4 horizons × 24 regime-stratified sites (52 models), scoring every model against both persistence and a leakage-controlled day-of-year climatology. The two baselines order the same models differently, and the disagreement is a systematic function of horizon rather than noise.
2. A **feature ablation that separates the two questions** the comparison is easily confused between. Losing to climatology does *not* imply seasonal skill: of the 11 models that beat persistence and lose to climatology, 10 retain their skill with every calendar feature removed, while the most seasonally dependent variable in the set comfortably *beats* climatology. Diagnosing mechanism needs an ablation; the baseline comparison answers a different question.
3. **Leave-one-site-out spatial validation** blocked in space and time simultaneously (1248 held-out fits), testing the assumption that licenses serving a pooled model at coordinates it was not fitted on — and finding that transfer is partial, with the median skill falling roughly threefold.

4. **Negative results reported in full**, including 5 model–horizon combinations that lose to persistence outright. These identify a class of high-inertia variables for which the approach is contraindicated.
5. A **practical decision procedure** (Fig. 7) mapping the pair $(S_{\text{pers}}, S_{\text{clim}})$ to a deployment recommendation.

2 Related Work

Skill scores relative to a reference forecast are standard in numerical weather prediction, and are conventionally reported against climatology, persistence, or both [1,4]. The concern raised here is not that climatology is unknown as a baseline — it is long-established in NWP — but that in the recent machine-learning-for-ocean literature persistence is frequently the sole reference, and the specific interaction between baseline choice and forecast horizon is rarely made explicit.

Data-driven ocean and atmospheric forecasting has moved rapidly from convolutional recurrent architectures for sea surface temperature [7] to global neural weather models [5,3]. Evaluation practice has not kept pace uniformly: gains are often reported for a single variable at a single horizon against a single baseline. Our contribution is orthogonal to architecture. We deliberately hold the model class fixed — gradient-boosted trees, one configuration, no variable-specific code — so that variation in the results is attributable to the variable, the horizon and the baseline rather than to model tuning.

The leakage discipline we adopt for the climatology follows standard practice in ecological and environmental modelling, where fitting any component of a baseline or label definition on the evaluation period is a known and consequential error [6]. Our spatial validation follows the same literature's argument against random k-fold splits on spatially autocorrelated data.

3 System Architecture

Figure 1 shows the framework. It is deliberately uniform: one acquisition path, one feature builder and one model class serve every variable, and nothing in the pipeline branches on which variable it is training. Adding a variable is a configuration entry, not a code change. This uniformity is what makes a cross-variable audit meaningful — differences in the results cannot be attributed to differences in the implementation.

4 Data and Methods

4.1 Variables and sites

We study 13 ocean variables spanning physics, surface waves and biogeochemistry (Table 1), all drawn from operational products: Copernicus Marine Service

Table 1. The 13 variables audited. "Feature rows" is the pooled count across all 24 sites after feature construction and target alignment.

Variable	Unit	Sites	Feature rows
bottom temperature	degC	24	18,960
chlorophyll a	mg/m3	24	18,960
dissolved oxygen	mmol/m3	24	18,960
maximum wave height	m	24	18,888
mean wave period	s	24	18,888
nitrate	mmol/m3	24	18,960
ph	pH	24	18,960
primary productivity	mg/m3/day	24	18,960
sea level anomaly	m	24	10,200
significant wave height	m	24	18,888
water salinity	PSU	24	18,960
water temperature	degC	24	18,960
wave direction	deg	24	18,888

analysis and forecast fields for the physical and biogeochemical variables, and the same provider's wave products for the wave variables. Variables were included on the criterion that a complete 24-site history was available in our archive at the snapshot date — a data-availability criterion applied before any model was scored, discussed further in Section 7.

Each variable is sampled at 24 sites chosen to span the major ocean basins and dynamical regimes: western boundary currents (Gulf Stream, Kuroshio, Agulhas), eastern boundary upwelling systems (Peru, Benguela, Canary), subtropical gyres, marginal and shelf seas, and the Southern Ocean. Site locations are shown in Fig. 6. The record spans 2024-06-11 to 2026-08-09 at daily resolution.

4.2 Forecast framework

For a target variable y at horizon h, features at time t comprise lagged values of y and its covariates at lags $\{1, 3, 7, 14, 30\}$ days; rolling mean, standard deviation, minimum and maximum over windows of $\{7, 14, 30\}$ days; first differences; linear trend coefficients over $\{7, 30\}$ days; calendar and cyclical encodings of day of year; and static site descriptors (latitude, longitude, ocean depth, basin). All rolling windows close at t inclusive, which is safe because $h \geq 1$ is enforced. Exactly one forward shift exists in the pipeline, in target construction.

Models are LightGBM regressors [2] with fixed hyperparameters across all variables and horizons (400 trees, learning rate 0.05, 31 leaves). No per-variable tuning was performed: tuning would confound the cross-variable comparison that is the point of the audit.

Delta targets. Models regress on the *change* $y(t+h) - y(t)$ rather than the level $y(t+h)$. A regression tree partitions feature space into piecewise-constant regions and therefore cannot represent the identity function $y(t+h) = y(t)$ — which is exactly what persistence is. Fitted on levels, the framework scored worse than persistence at every horizon tested. Under a delta target persistence becomes the constant zero function, which a tree represents exactly, and the same data and features yield positive skill. Predictions are decoded by adding the model output to the anchor $y(t)$. For circular variables (wave direction) the target is the signed angular change wrapped to $(-180°, 180°]$ and decoding is modulo $360°$.

Pooled models, and what "pooled" does and does not mean. One model per (variable, horizon) is fitted across all 24 sites jointly, with site coordinates and depth as features, rather than one model per site. We call this *pooled* rather than *global*, and the distinction is worth stating explicitly because the two are easy to conflate.

Pooling is a modelling choice: it lets one model be evaluated at a coordinate it was not fitted on, which is what makes the spatial question in Section 5.5 askable at all. It is *not* a claim of global coverage. 24 sites do not sample the world ocean; they were selected to span distinct dynamical regimes (Section 4.1), which makes them a regime-stratified sample rather than a representative one. Any statement here about behaviour at unseen coordinates is therefore evidence about *these* regimes, and Section 5.5 reports how far even that extends.

4.3 Baselines

Persistence. $\hat{y}_{\text{pers}}(t+h) = y(t)$, the last observed value at feature time. This is also the anchor of the delta target, so a delta model that outputs zero *is* persistence exactly.

Climatology. For each site s and day of year d, the climatology is the mean of all training observations at that site whose day of year falls within ± 15 days of d, treating the year as circular so that 31 December neighbours 1 January:

$$\hat{y}_{\text{clim}}(s, d) = \frac{1}{|W(d)|} \sum_{t \in W(d)} y_s(t), \tag{1}$$

where $W(d) = \{t : \delta(\text{doy}(t), d) \leq 15\}$ and δ is circular day-of-year distance. The window is necessary rather than cosmetic: the record is roughly 2.2 years, so a bare (site, day-of-year) mean would average two samples and be mostly noise — a strawman the model would beat for the wrong reason. A ± 15-day window pools approximately 60 samples per estimate. For circular variables the mean is the circular mean, computed via atan2 of the mean sine and cosine; the arithmetic mean of $359°$ and $1°$ is $180°$, which would make the baseline point in exactly the wrong direction.

Leakage control. The climatology is refitted *inside every cross-validation fold* on that fold's training rows only (shaded in Fig. 2). Fitting a climatology once over the full record would leak the evaluation period into the baseline's own definition, flattering the baseline and understating the model — an error in the "safe" direction, and therefore one that is easy to leave in place. Fit and application are implemented as two separate functions so that fitting on the wrong rows requires deliberate effort, and the discipline is enforced by a unit test that shifts the second year of a synthetic series by a known constant and asserts the baseline's error on that period recovers the shift.

4.4 Validation

Figure 2 shows both experiments end to end.

Rolling-origin cross-validation. Five expanding-window folds, ordered in time, with an embargo gap equal to the forecast horizon between each training window and its test window — a training row within h days of the test window shares its target period. Random k-fold splitting is never used. The pooled multi-site frame is sorted by time so that a fold boundary separates *dates*, and no site's future can leak into another site's past.

Skill score. For a reference forecast r,

$$S_r = 1 - \frac{\mathrm{RMSE}^2_{\mathrm{model}}}{\mathrm{RMSE}^2_r}, \tag{2}$$

so $S_r > 0$ means the model beats reference r, $S_r = 0$ is parity, and $S_r < 0$ means the reference is better. We report S_{pers} and S_{clim} for every model. Where the climatology cannot answer for some test rows (a day of year no site in the training fold observed), both model and baseline are scored on the intersection, and the reduced count is reported rather than the rows being silently dropped.

4.5 Reproducibility

All experiments run from a fixed archival snapshot (2026-08-10), pinned so that window boundaries — and therefore every cached provider request — are identical across reruns. Without this pinning each rerun re-requests the record from upstream providers and, because those providers intermittently return partial responses, occasionally scores a slightly different dataset, making two runs of the same experiment non-comparable. Every figure and table cell in this paper is generated programmatically from the experiment output; no number is transcribed by hand.

5 Results

5.1 The two baselines disagree, and disagree with horizon

Figure 3 is the central result. For each variable it plots skill against persistence and skill against climatology as a function of horizon.

Table 2. Skill against persistence (S_p) and against climatology (S_c) at each horizon. Bold marks a negative score — the baseline winning. Reading across a row shows the divergence; reading down the S_c columns shows it is systematic.

	$h=1$		$h=3$		$h=7$		$h=30$	
Variable	S_p	S_c	S_p	S_c	S_p	S_c	S_p	S_c
bottom temperature	0.253	0.992	**-0.033**	0.935	**-0.032**	0.823	**-0.109**	0.502
chlorophyll a	0.530	0.969	0.190	0.729	0.220	0.400	0.272	0.192
dissolved oxygen	0.377	0.957	0.190	0.723	0.230	0.410	0.443	0.020
maximum wave height	0.249	0.572	0.327	0.047	0.362	**-0.042**	0.386	**-0.100**
mean wave period	0.230	0.561	0.310	0.043	0.374	**-0.026**	0.429	**-0.055**
nitrate	0.404	0.974	0.090	0.777	0.098	0.459	0.273	**-0.051**
ph	0.441	0.975	0.154	0.792	0.129	0.481	0.297	**-0.148**
primary productivity	0.284	0.927	0.145	0.646	0.186	0.316	0.178	0.026
sea level anomaly	0.402	0.993	0.407	0.961	0.082	0.770	0.068	0.449
significant wave height	0.272	0.580	0.331	0.045	0.360	**-0.044**	0.379	**-0.108**
water salinity	0.050	0.958	**-0.168**	0.753	**-0.089**	0.480	0.100	**-0.023**
water temperature	0.075	0.958	0.009	0.817	0.139	0.633	0.430	0.187
wave direction	0.090	0.445	0.112	0.029	0.151	**-0.157**	0.163	**-0.254**

The two curves systematically diverge. Median skill against persistence is 0.272 at 1 d and 0.273 at 30 d — essentially unchanged, having dipped to 0.154 and 0.151 in between. Over exactly the same range, median skill against climatology falls monotonically: 0.958, 0.729, 0.410, -0.023. The fall is not a property of a few variables — climatology skill is lower at 30 d than at 1 d for 13 of 13 variables, without exception.

This is the structural prediction of Section 1 realised in data. Persistence becomes easy to beat as the horizon grows, which holds the persistence-referenced score up even as the models' own errors grow; climatology becomes *harder* to beat, because the seasonal cycle is an increasing share of what remains predictable at all.

The practical consequence is that the two baselines license opposite statements about the same models. Reported against persistence alone, this framework appears *horizon-invariant* — as skilful at a month as at a day, which for a forecasting system is a remarkable claim. Reported against climatology, the median model has fallen *below* the seasonal cycle by 30 d (-0.023), and 7 of 13 variables lose to it outright there, against 1 that lose to persistence. Both readings come from the same predictions. Per-variable scores are in Table 2.

5.2 What survives the seasonal cycle

The consequential cases are the 11 model–horizon combinations with $S_\mathrm{clim} < 0$ while $S_\mathrm{pers} > 0$. These models beat the last observed value and lose to the seasonal cycle. Under the reporting convention this literature most often uses, every one of them would be presented as a success — and for each, the correct operational decision is to ship the climatology instead, which is cheaper to compute, cheaper to serve, and more accurate.

It is tempting to conclude from this that the models' long-horizon skill *is* seasonal — that they have learned little beyond the time of year. That inference does not follow, and Section 5.3 shows it is mostly wrong.

5.3 Is the long-horizon skill seasonal? A feature ablation

Losing to climatology shows a model is *no better than* the seasonal cycle. It does not show the model's skill *comes from* the seasonal cycle. Those are different claims, and only the second is an explanation. We test it directly with three arms over the same 52 variable–horizon pairs, sharing identical rows, folds and hyperparameters, differing only in the feature set:

- **full** — every feature the shipped model uses.
- **no calendar** — the same, minus day-of-year, month, week and their sine/cosine encodings. If skill survives this, it was not seasonal.
- **calendar only** — calendar plus static site descriptors, and nothing else: no lags, no rolling statistics, no trends, and not the raw contemporaneous value $y(t)$ either. This is the ceiling on a purely seasonal model within this framework.

The exclusion of $y(t)$ from the third arm matters. The feature builder retains it deliberately — it is the strongest short-horizon predictor available — so an arm that merely zeroed the lag lists would still contain it, and would be a persistence model wearing a seasonal label.

Removing the calendar costs very little. At 30 d the median model retains 0.76 of its skill with no calendar information at all, and 10 of 12 variables still beat persistence without it. At 1 d the median retention is 0.97.

A purely seasonal model is not competitive. The calendar-only arm beats persistence in just 3 of 52 cases, with a median skill of -0.172. Calendar and location alone cannot forecast these variables.

The models that lose to climatology are not the seasonal ones. This is the result that overturns the natural reading. Of the 11 model–horizon combinations that beat persistence while losing to climatology, 10 remain positive against persistence with the calendar removed, retaining a median 0.86 of their skill. Their skill is not seasonal in origin. They are simply not as accurate as the seasonal cycle — which is a statement about the climatology's strength, not about what the model learned.

Conversely, of the 6 cases where the calendar does carry more than half the skill, 5 *beat* climatology. The clearest genuine seasonal case is water temperature at 30 d: removing the calendar drops it from 0.430 to 0.139 (a 0.68 share), and the calendar-only arm reaches 0.281 on its own — yet its skill against climatology is 0.187, comfortably positive. A reader applying the climatology test alone would have classified this variable as the healthy one.

Table 3. Skill against persistence under each feature arm at 30 d, with skill against climatology for reference. Bold marks the cases where removing the calendar destroys more than half the skill — the genuinely seasonal ones, which are a minority and are mostly *not* the ones that lose to climatology.

Variable	Full	No calendar	Calendar only	S_{clim}
bottom temperature	-0.109	-0.126	-0.476	0.502
chlorophyll a	0.272	0.239	-0.483	0.192
dissolved oxygen	0.443	0.373	0.124	0.020
maximum wave height	0.386	0.301	-0.117	-0.100
mean wave period	0.429	0.385	-0.149	-0.055
nitrate	0.273	0.257	-0.386	-0.051
ph	0.297	0.196	0.071	-0.148
primary productivity	0.178	0.132	-0.745	0.026
sea level anomaly	0.068	**-0.019**	-0.727	0.449
significant wave height	0.379	0.297	-0.117	-0.108
water salinity	0.100	**-0.087**	-0.649	-0.023
water temperature	0.430	**0.139**	0.281	0.187
wave direction	0.163	0.105	-0.012	-0.254

Consequence for practice. S_{clim} and seasonal dependence are close to orthogonal here (Fig. 4b). Comparing against climatology answers “should I ship this model, or the climatology?” — an operational question, and the comparison answers it correctly. It does not answer “what has this model learned?”. Answering that needs an ablation, which costs one extra training run per configuration and is the only one of our three diagnostics that addresses mechanism at all. Table 3 reports all three arms.

5.4 Where the framework fails

5 model–horizon combinations have $S_{\mathrm{pers}} < 0$: the model is worse than doing nothing. These concentrate in the high-inertia subsurface variables, and the pattern is physically coherent. For a variable whose autocorrelation time greatly exceeds the forecast horizon, persistence is close to optimal and the residual the delta model is asked to predict is close to pure noise; fitting it adds variance without adding signal. This is a property of the variable rather than a defect of the implementation, and it is useful precisely because it is predictable in advance from the autocorrelation structure — it identifies where this class of model should not be deployed at all.

Notably these are not the same variables that fail against climatology. Some subsurface variables lose to persistence while beating climatology by a wide margin; some wave variables do the reverse. The two baselines fail in different places, which is the strongest available argument for reporting both.

Table 4. Leave-one-site-out results. "Med." is the median skill against persistence across the 24 held-out sites; n^+ counts sites retaining positive skill. Bold marks a negative median — a variable that does not transfer.

	$h = 1$		$h = 3$		$h = 7$		$h = 30$	
Variable	Med.	n^+	Med.	n^+	Med.	n^+	Med.	n^+
bottom temperature	0.23	16	**-0.17**	10	**-0.45**	3	**-3.02**	4
chlorophyll a	0.37	21	0.05	13	**-0.16**	9	**-0.16**	10
dissolved oxygen	0.44	24	0.17	17	0.18	16	0.27	17
maximum wave height	0.21	23	0.23	23	0.32	22	0.40	23
mean wave period	0.17	23	0.26	21	0.36	22	0.41	23
nitrate	0.11	13	**-0.54**	6	**-2.85**	6	**-1.18**	8
ph	0.45	24	0.12	21	0.08	14	0.25	17
primary productivity	0.17	16	0.01	12	0.05	13	0.01	13
sea level anomaly	**-0.08**	10	0.01	13	**-0.11**	8	**-0.03**	10
significant wave height	0.21	23	0.21	21	0.27	21	0.36	23
water salinity	0.14	16	**-0.03**	10	**-0.17**	6	**-0.04**	10
water temperature	0.13	18	0.06	14	0.11	16	0.52	20
wave direction	0.12	18	0.12	16	0.15	20	0.24	20

5.5 Spatial transfer: leave-one-site-out

Serving a pooled model at coordinates it was not fitted on rests on an assumption that is rarely tested: that the model generalises to ocean it never saw during training, rather than interpolating among its training sites. We test this directly. For each variable, horizon and site, we refit on the remaining $24 - 1$ sites and score on the held-out one (1248 fits).

The split is blocked in space *and* time. Training uses the earlier 80% of the record at the 23 retained sites; testing uses the later 20% at the held-out site. A purely spatial hold-out would leave the model trained on the same dates it is tested on, measuring interpolation across a shared period rather than transfer. The climatology baseline is held to the same standard: it is fitted only on the 23 training sites, so for the held-out site it falls back to the nearest fitted site by great-circle distance — the spatial analogue of what the model itself is being asked to do.

Transfer is real but costs skill. At 7 d, median skill at held-out sites is 0.050 against 0.151 pooled, and 56.4% of held-out site-fits retain positive skill. That the majority remain positive supports serving the pooled model at unseen coordinates; that the median falls means the pooled figure overstates what a user clicking on open ocean should expect, and a deployed system should say so. Per-variable summaries are in Table 4.

Figure 6 shows transfer is not spatially uniform. Sites in dynamically energetic regions — western boundary currents and upwelling systems — are systematically harder to predict from elsewhere, the expected result: their variability is driven by mesoscale processes with short decorrelation scales that a model fitted on distant sites cannot resolve.

6 Discussion

6.1 A decision procedure

Figure 7 states the practical consequence of these results as a procedure. The pair $(S_{\text{pers}}, S_{\text{clim}})$, evaluated across more than one horizon and checked against a held-out site, maps onto a small number of clearly distinguishable recommendations — three of which are "do not deploy this model" for reasons that a single baseline cannot surface.

6.2 Implications

Report both baselines. Persistence and climatology are individually cheap and jointly diagnostic, and their disagreement carries information neither provides alone. A model that beats persistence at long range while losing to climatology is one a persistence-only convention would report as a success, when the cheaper baseline is more accurate. That is a real and consequential failure mode, and it fails in the flattering direction.

But do not over-read the climatology comparison. We initially read those cases as evidence that the models had learned little beyond the time of year. The ablation in Section 5.3 shows that reading is wrong for 10 of the 11 cases it was proposed to explain: their skill survives the calendar being removed. Losing to climatology is a statement about the *baseline's* strength, not about the model's mechanism. We report this because the incorrect inference is the natural one, it is rhetorically attractive, and nothing in a two-baseline comparison rules it out — which is precisely why a third diagnostic is needed.

Baseline choice interacts with horizon, so single-horizon evaluation hides it. The divergence in Fig. 3 is only visible across horizons. A paper reporting one horizon — however carefully — can report a number that is correct and still leave the reader unable to tell which of two very different models produced it.

Architecture is not the lever these results point to. We held the model class fixed deliberately. Nothing here argues that gradient-boosted trees are the right architecture; the results argue that *evaluation* is currently the binding constraint on interpreting such comparisons. A more expressive architecture evaluated against persistence alone at a single horizon would produce a number that is neither comparable to these nor obviously more informative.

Pooling across sites is defensible, within limits that should be stated. The leave-one-site-out results support serving a pooled model at coordinates it was not fitted on — but only just, and only within the regimes sampled here. A median held-out skill of 0.050 against 0.151 pooled, with 56.4% of held-out sites retaining positive skill, is evidence of partial transfer, not of a model that can be pointed anywhere. Two qualifications follow, and neither is rhetorical.

First, the pooled figure *is* the number a deployed system would quote, and it overstates what a user selecting an unsampled location should expect — by roughly a factor of three at 7 d in the median. Reporting the held-out figure instead is the honest choice.

Second, 24 sites spanning selected regimes cannot establish behaviour over the world ocean, and we do not claim it. What the experiment establishes is that transfer degrades, that the degradation is largest in the dynamically energetic regions users are most likely to care about, and that the direction of the error is optimistic. Extending this to a genuine statement about global coverage would need a site set an order of magnitude larger, drawn to represent rather than to stratify.

7 Limitations

Record length. The archive spans roughly 2.2 years. This is the most consequential limitation and it bears directly on the central comparison: a climatology estimated from two annual cycles is weakly constrained, and the ±15-day window is what makes it a serious baseline rather than a noisy one. A longer record would strengthen the climatology and, we expect, *reduce* the measured S_{clim} — our headline comparison is therefore conservative in the direction that favours the model, and the qualitative conclusion is unlikely to reverse. It also means we cannot separate the seasonal cycle from interannual variability.

Variable coverage. The 13 variables reported are those with complete 24-site coverage in the archive at the snapshot date. Others were excluded on data availability, not on their results. The criterion was applied before any model was scored, so it does not select on outcome, but the set is not a random sample of ocean variables — it is skewed toward well-observed operational products.

Spatial coverage, and what "pooled" is not. 24 sites are not a sample of the world ocean. They were chosen to span distinct dynamical regimes — western boundary currents, eastern boundary upwelling systems, subtropical gyres, marginal seas, the Southern Ocean — which makes them regime-stratified rather than representative, and deliberately over-weights energetic regions relative to their true area. Two consequences follow. Aggregate statistics across sites are averages over regimes, not area-weighted ocean averages, and should not be read as the latter. And the leave-one-site-out result establishes that transfer degrades *between these regimes*; it does not establish behaviour at an arbitrary point on the globe, which would need a site set an order of magnitude larger drawn to represent rather than to stratify. We use "pooled" throughout for the modelling choice of fitting one model across sites, and avoid "global", precisely because the two are easy to conflate and only the first is established here.

Reanalysis as ground truth. The "observations" are operational analysis products, not in-situ measurements. They are themselves model output constrained by assimilated observations, so our scores measure agreement with an analysis rather than with the ocean. This is standard for work at this spatial coverage, but it bounds what the results can claim, particularly in sparsely observed regions where the analysis is least constrained.

Circular variables are modelled linearly. Wave direction is decoded circularly and scored with circular error, but the underlying regression treats the wrapped delta as a linear target: a veer of 5° across north trains as $-355°$. The principled fix — predicting vector components and deriving the angle — is not applied here, and the direction results should be read with that caveat.

Fixed hyperparameters. No per-variable tuning was performed. This is necessary for the cross-variable comparison but means every individual model is likely below its own achievable ceiling. The audit measures the framework as configured, not the best attainable skill per variable.

8 Conclusion

Skill against persistence and skill against climatology are not interchangeable summaries of the same quantity, and the difference between them is not noise: it is a systematic function of forecast horizon that can invert the qualitative conclusion. Across 13 ocean variables, 24 sites and four horizons, median skill against persistence is no lower at a month than at a day, while median skill against climatology falls monotonically to -0.023 — declining with horizon for 13 of 13 variables without exception. 11 model–horizon combinations beat persistence while losing to the seasonal cycle outright.

The natural explanation for those cases — that the models had learned little beyond the time of year — is one we proposed and then had to withdraw. A feature ablation shows 10 of the 11 keep their skill with the calendar removed, and that the most seasonally dependent variable in the study beats climatology rather than losing to it. Two baselines tell you which forecast to ship; they do not tell you what the model learned, and we could not have established the difference without the third experiment.

We also find that pooled models transfer only partially to sites they never saw: 56.4% of held-out sites retain positive skill at 7 d, but the median falls from 0.151 to 0.050 — a cost pooled metrics conceal, concentrated in dynamically energetic regions. Across 24 regime-stratified sites this is evidence about those regimes, not about the world ocean.

The recommendation is cheap to adopt: report both baselines, across more than one horizon, and report the cases where the baseline wins.

Reproducibility. Experiments run from a pinned archival snapshot (2026-08-10). The climatology implementation, its leakage tests, the experiment driver and the asset generator that produces every figure and table are included in the accompanying source tree (`forecasting/climatology.py`, `tests/test_climatology.py`, `scripts/run_paper_experiments.py`, `scripts/build_paper_assets.py`).

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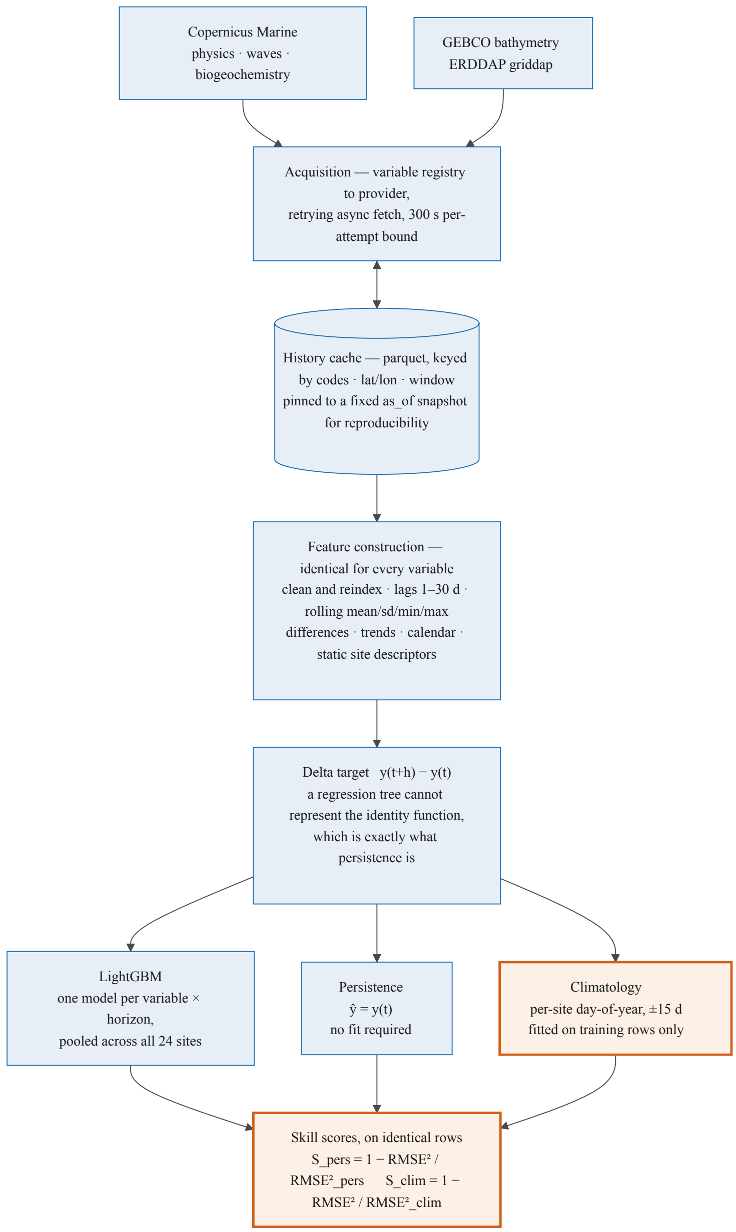

Fig. 1. Framework architecture. One acquisition path, one feature builder and one model class serve all 13 variables. The three predictors at the bottom are scored on identical rows. The contribution of this paper is shaded: the climatology baseline and the second skill score it makes possible.

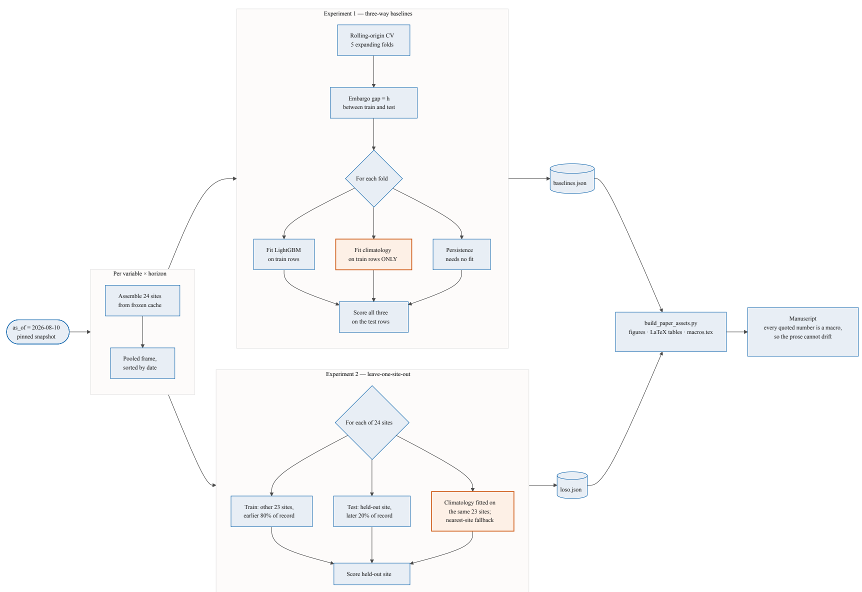

Fig. 2. Experiment data flow. A pinned `as_of` snapshot makes every provider request cache-stable, so the experiments are reproducible and run offline. Shaded boxes are the two places a baseline is fitted, and both are fitted strictly inside the fold they are scored in.

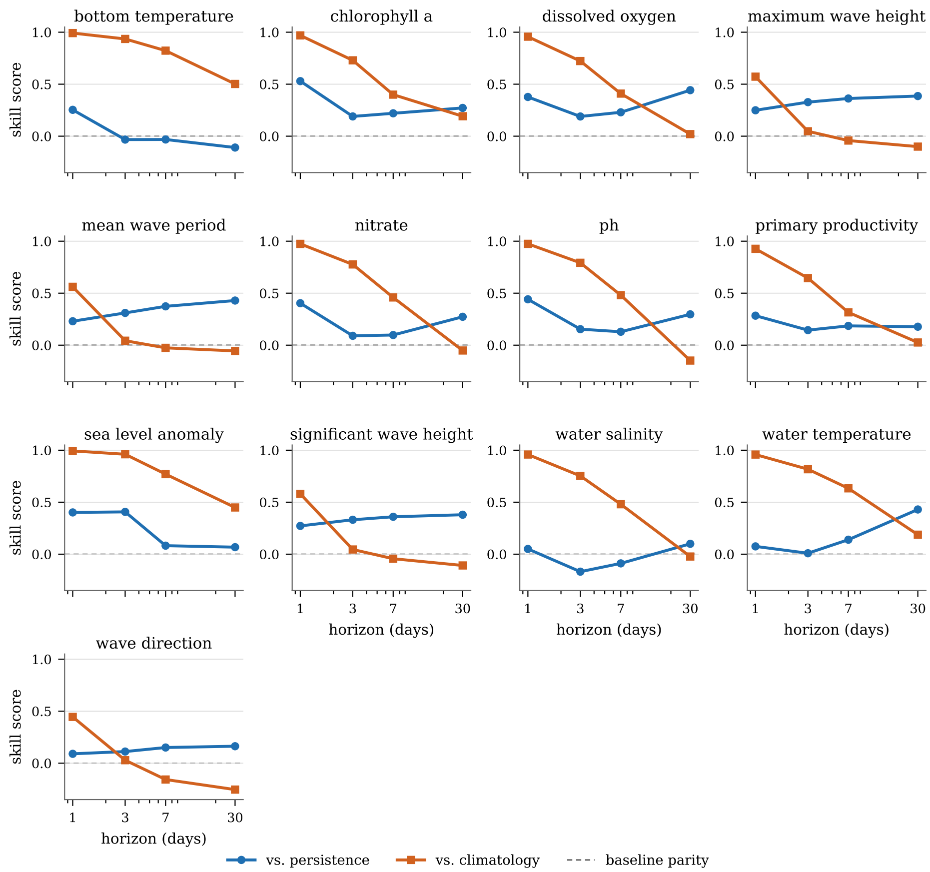

Fig. 3. Skill against two baselines, by forecast horizon, for each of the 13 variables. The climatology curve (orange squares) falls in every panel, without exception; the persistence curve (blue circles) is flat or rising in most. The dashed line is parity; points below it mean the baseline wins. A reader given only the blue curve would conclude these models hold their skill out to a month.

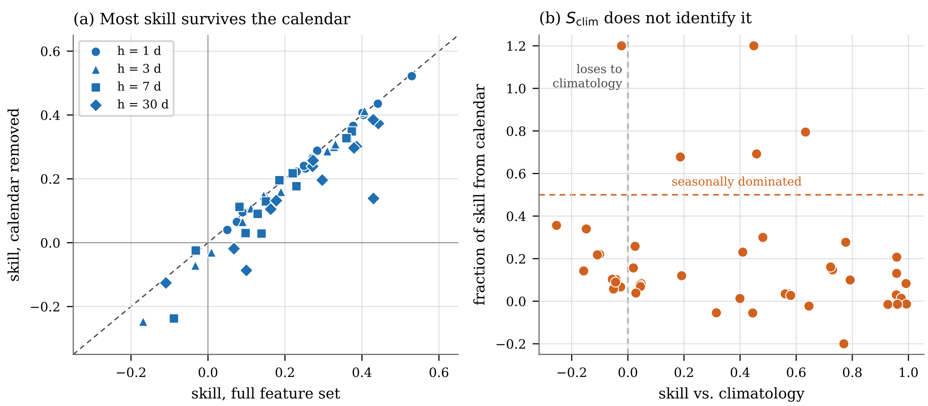

Fig. 4. (a) Skill with the full feature set against skill with every calendar feature removed; points on the diagonal lost nothing. (b) The fraction of skill attributable to the calendar, against skill versus climatology. If losing to climatology indicated seasonal skill, the points left of zero would sit high. They do not.

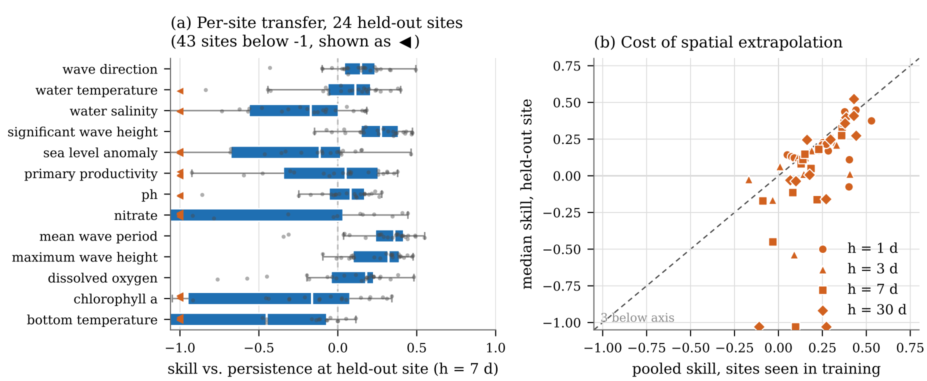

Fig. 5. Leave-one-site-out transfer. (a) Skill at each held-out site (7 d); each dot is one site scored by a model that never saw it. (b) Pooled skill at sites seen in training against median skill at held-out sites; the dashed line is equality, and points below it are the cost of spatial extrapolation.

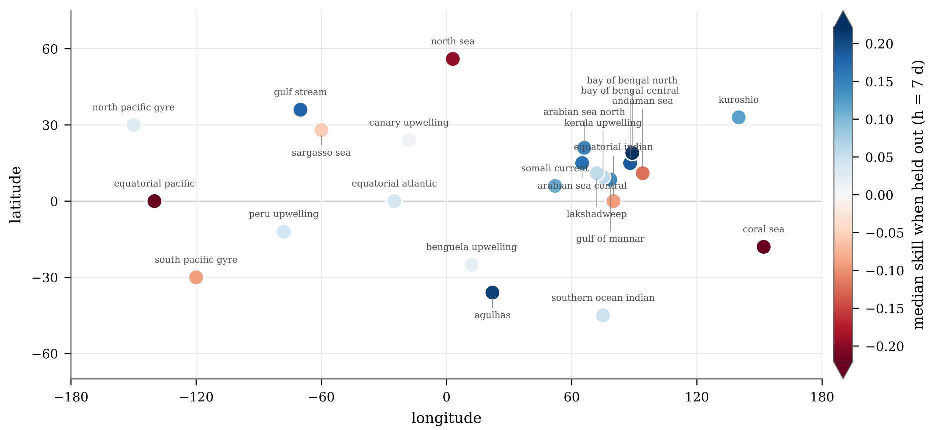

Fig. 6. The 24 evaluation sites, coloured by median skill across variables when that site is held out entirely (7 d). Blue indicates the model transfers there; red indicates persistence wins. The colour bar is clipped at a robust bound so that one catastrophic failure does not compress the rest of the scale.

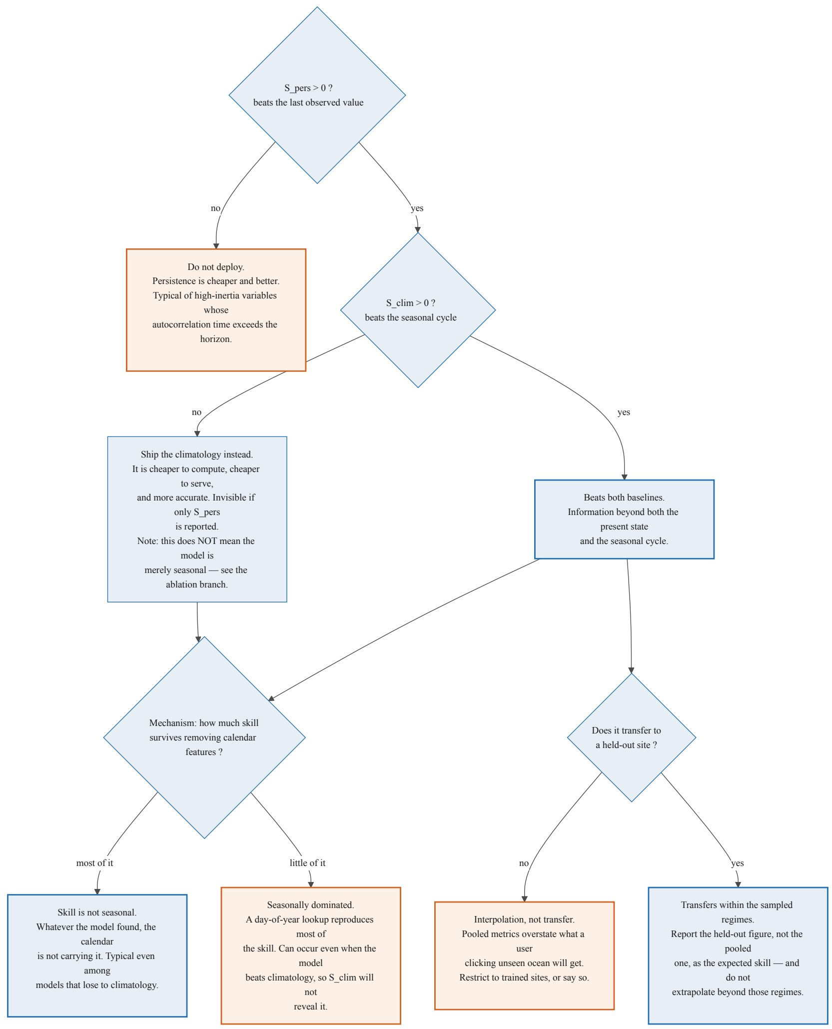

Fig. 7. Interpreting the two skill scores together. Three of the six terminal states are negative recommendations that reporting S_{pers} alone would not reveal.